

HDOC Day-37

Menu-Driven String Operations in C

Aim

Write a menu-driven C program using separate functions to: convert a lowercase string to uppercase without built-in case-conversion functions, reverse a string, check whether a string is a palindrome, and exit.

Algorithm

1. Start the program and display a menu with four choices.
2. Read the user's choice.
3. For choice 1, read a string and convert each lowercase letter by subtracting 32 from its ASCII value.
4. For choice 2, read a string, find its length, and print characters from the last position to the first.
5. For choice 3, compare characters from both ends of the string. If all corresponding characters match, the string is a palindrome.
6. For choice 4, terminate the program.
7. Repeat the menu until Exit is selected.

C Program

```
#include <stdio.h>

void toUpper(char s[]) {
    int i;
    for (i = 0; s[i] != '\0'; i++)
        if (s[i] >= 'a' && s[i] <= 'z')
            s[i] = s[i] - 32;
}

int length(char s[]) {
    int i = 0;
    while (s[i] != '\0') i++;
    return i;
}

void reverse(char s[]) {
    int i, n = length(s);
    for (i = n - 1; i >= 0; i--)
        printf("%c", s[i]);
    printf("\n");
}

int palindrome(char s[]) {
    int i, n = length(s);
    for (i = 0; i < n / 2; i++)
        if (s[i] != s[n - 1 - i])
            return 0;
    return 1;
}

int main() {
    int choice;
    char str[100];

    do {
        printf("\n1. Convert lowercase to uppercase");
        printf("\n2. Reverse a string");
        printf("\n3. Check palindrome");
        printf("\n4. Exit");
```

```

printf("\nEnter choice: ");
scanf("%d", &choice);

if (choice >= 1 && choice <= 3) {
    printf("Enter string: ");
    scanf(" %99[^\n]", str);
}

switch (choice) {
    case 1:
        toUpper(str);
        printf("Uppercase: %s\n", str);
        break;
    case 2:
        printf("Reversed: ");
        reverse(str);
        break;
    case 3:
        if (palindrome(str))
            printf("Palindrome\n");
        else
            printf("Not a palindrome\n");
        break;
    case 4:
        printf("Exiting...\n");
        break;
    default:
        printf("Invalid choice\n");
}
} while (choice != 4);

return 0;
}

```

Sample Test Cases

Choice	Input	Expected Result
1	hello world	HELLO WORLD
2	computer	retupmoc
3	madam	Palindrome
3	hello	Not a palindrome
4	-	Program exits

Result

The menu-driven C program performs uppercase conversion, string reversal, and palindrome checking using separate functions.

Kali Linux Terminal - Test Run 1

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ vi day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ cc day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ ./a.out
```

1. Convert lowercase to uppercase
2. Reverse a string
3. Check palindrome
4. Exit

Enter choice: 1

Enter string: hello world

Uppercase: HELLO WORLD

Kali Linux Terminal - Test Run 2

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ vi day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ cc day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ ./a.out
```

1. Convert lowercase to uppercase
2. Reverse a string
3. Check palindrome
4. Exit

Enter choice: 2

Enter string: computer

Reversed: retupmoc

Kali Linux Terminal - Test Run 3

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ vi day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ cc day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ ./a.out
```

```
1. Convert lowercase to uppercase
```

```
2. Reverse a string
```

```
3. Check palindrome
```

```
4. Exit
```

```
Enter choice: 3
```

```
Enter string: madam
```

```
Palindrome
```

Kali Linux Terminal - Test Run 4

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ vi day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ cc day37.c
```

```
(kali~kali)-[~/Desktop/cprog]
```

```
└─$ ./a.out
```

```
1. Convert lowercase to uppercase
```

```
2. Reverse a string
```

```
3. Check palindrome
```

```
4. Exit
```

```
Enter choice: 3
```

```
Enter string: hello
```

```
Not a palindrome
```